APPENDIX II—CAPE VERDE: DEBT SUSTAINABILITY ANALYSIS

Cape Verde's debt remains sustainable, but risks have increased since the previous debt sustainability analysis (DSA) in November 2010. *A faster accumulation of external public debt over the next few years combined with a deteriorating global economic environment accentuates macroeconomic vulnerabilities. Improvement of the debt outlook requires faster reduction of the fiscal deficit and implementation of long-standing structural reforms to boost growth. New public investment projects should directly support private sector investment and export growth.*

I. BACKGROUND

1. **This DSA updates an earlier DSA, covering 2010–30 (IMF Country Report No. 10/349).** This earlier DSA found that public debt levels had increased as external borrowing was scaled up to finance public investment. The PV of external debt-to-GDP ratio was projected to approach the risk threshold in 2012 under the baseline, and to breach some stress tests. Debt service ratios remained below their thresholds, and resilient to the stress tests. The overall debt sustainability outlook was assessed as broadly unchanged compared to the previous DSA in 2008 (IMF Country Report No. 08/37).

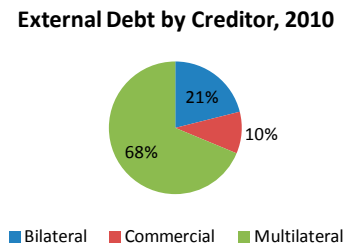
2. **This DSA reflects information available as of mid-October 2011 and uses the stock information at end-2010.** The debt data currently covers central government external and domestic debt. External debt contracted by the central government on behalf of state-owned enterprises is also included. Publically-guaranteed debt extended to state-owned enterprises and local governments is included, but coverage is still hampered by data limitations. Technical assistance from the World Bank and the IMF during 2011 supported the authorities in improving measurement of the external debt stock and the associated projection of repayment flows. Both institutions assisted the authorities in the preparation of the Medium-Term Debt Strategy finalized in 2011.

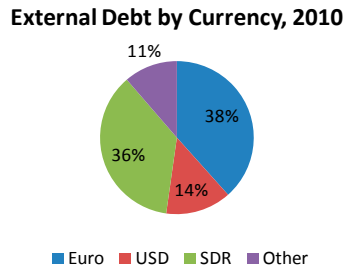
3. **Public debt increased in 2010 as the government implemented its public investment program.** The nominal stock of public debt rose to 74.9 percent of GDP in 2010 from 68.8 percent in 2009[1] The rise reflects a high level of public investment, which increased to 18.7 percent of GDP in 2010 from 14.0 percent in 2009. Total revenue worsened by around one percentage point of GDP during the period, but it was more than offset by a reduction in government current expenditures.

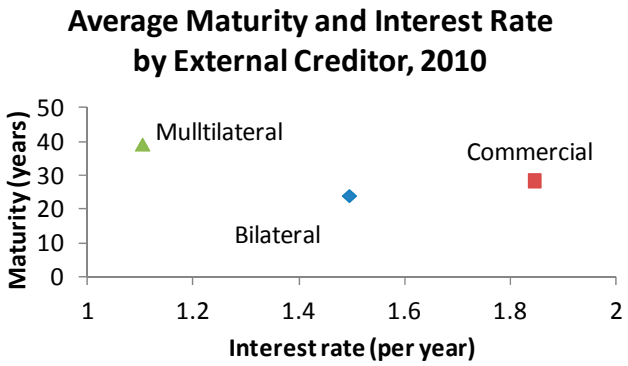
[1] The latest official nominal GDP estimates in Cape Verde date from 2007. The GDP estimates between 2008 and 2010 used in the DSA were constructed by the staff and are broadly aligned with the Ministry of Finance estimates. However, the BCV's GDP estimates are lower by about 10 percent for 2010, mainly reflecting differences in the deflator estimates.

4. **External public debt is mostly concessional.** External public debt to GDP reached 51.3 percent in 2010 (Table 1). Multilateral institutions, and in particular, the World Bank Group and the African Development Bank, are the main external creditors (Text Figure).[2] The portfolio has a long maturity profile and a low average interest rate, and over 95.0 percent is contracted at a fixed rate. Considering that the U.S. dollar corresponds to 41.9 percent of the SDR valuation basket, approximately 30.0 percent of the external debt stock is sensitive to the U.S. dollar-euro exchange rate fluctuations.

Text Figure. Cape Verde: External Indicators







Source: Cape Verdean authorities.

[2] In December 2011, the African Development Bank disbursed a budget support loan in the amount of 15 million euros. This loan is on quasi-concessional basis and within the limit of non-concessional loans agreed in the context of the current PSI.

5. **External private debt poses a potential risk.** The authorities are compiling information about private sector external debt, but there is a need for a more systematic monitoring of actual disbursements and repayment flows. Based on the available limited information, staff estimates that external private debt was 20 percent of GDP at the end of 2010–70 percent of this total corresponded to nonbank debt. The peg has facilitated this high level of debt, and the authorities need to factor this into their management of the exchange rate regime.

6. **Domestic debt remains stable, at a manageable level.** The authorities have been able to contain domestic financing of the central government. Between 2009 and 2010, central government domestic debt remained at 23.5 percent of GDP. As of December 2009, the National Pension Fund (INPS) held approximately 35.0 percent of government domestic debt. The INPS has been running a surplus, on a current basis, and it is typically a key buyer of government debt in the primary market. The average interest rate on domestic debt is around 5.3 percent and average maturity is 7 years.

7. **From a consolidated perspective, staff estimates that the financial assets of the INPS appear to be larger than the liabilities of the state-owned enterprises and local governments.**[3] Guaranteed domestic debt of state-owned enterprises and municipalities amounted to an estimated 4.3 percent of GDP at end-2010.[4] The INPS financial assets amounted to an estimated 18.1 percent of GDP at end-2009. Therefore, consolidation of the public sector seems to point to an overall lower level of debt than looking at the central government alone.[5]

---

[3] Government-guaranteed external debt is included in the central government's debt stock. There may be some additional unguaranteed external debt, which staff are working to ascertain.

[4] Approximately 75 percent of the domestic debt guarantees pertain to the loss-making electricity company (Electra).

[5] Data limitations preclude a thorough consolidation of public debt beyond the central government. Financial information on the loss-making airline company (TACV) and municipalities is limited. The government is working with the World Bank and independent auditors to determine the financial situation of TACV, and is carrying out a Public Expenditure Management and Financial Accountability (PEMFAR) exercise with support from donors to determine the financial situation of municipalities. Also, an actuarial analysis of the future liabilities of the INPS needs to be carried out.

II. THE DSA BASELINE SCENARIO

8. **The DSA's macroeconomic assumptions reflect the negative spillovers from an uncertain external environment, and incorporate new medium-term fiscal projections.** Cape Verde's near-term growth outlook has worsened, reflecting the protracted and complex nature of the economic adjustment that lies ahead in Europe (Text Table). As a result, near-term growth is projected to remain below the annual average of 6.2 percent observed between 2000 and 2010. The main transmission channels of external uncertainty are foreign direct investment inflows and weaknesses in the real estate sector. Against this backdrop, the government announced an acceleration of the public investment program, leading to a higher deficit trajectory in the near term.[6]

Text Table. Cape Verde: Growth and Fiscal Assumptions

	2011–15	2016–31	2011–31
GDP growth (percent)			
Updated DSA	5.2	5.5	5.5
Previous DSA*	6.7	5.0	5.4
Fiscal balance (percent of GDP)			
Updated DSA	-7.7	-3.3	-4.3
Previous DSA*	-5.6	-2.5	-3.3

Sources: Cape Verdean authorities and IMF staff estimates.

* IMF Country Report No. 10/349.

9. **Long-term growth should benefit from gradual economic diversification.** Physical capital accumulation from tourism- and emigrant-related investments has been a key driver of economic growth in Cape Verde. The government is committed to deepening economic diversification through development of fisheries, the seahub, and the financial and communication sectors. The baseline assumes diversification to bear fruit at a gradual pace, and the annual average growth rate will remain below its historic average over the past decade.[7] This is in line with international experiences whereby economic diversification of small island states constitutes a significant development challenge.

10. **The largest projects of the public investment program are in infrastructure.** Almost half of the entire public capital spending is in the energy, water and transportation sectors. Deficiencies in infrastructure in these sectors constitute clear bottlenecks for private sector development. The public investment program also includes a social component,

[6] The public investment acceleration is being financed on concessional terms. Upon graduation from the least developed country (LDC) category in 2007, the international donor community agreed to provide a five-year window in which Cape Verde could still access concessional loans. Negotiations are on-going to extend the transition period until 2015. Cape Verde needs to reconsider its overall public investment program and the need now to build up fiscal buffers at a quicker pace.

[7] Staff is taking a relatively modest view about the growth dividends from the large infrastructure investment program, in light of the weak external environment.

human capital development, and sector-specific interventions to improve competitiveness (mainly in agriculture and fisheries).

11. **The baseline scenario is conditional on the government's commitment to low domestic financing.** The authorities have successfully controlled current expenditure to accommodate revenue shortfalls and public investment increases so far. Going forward, it is assumed that the government will continue to control fast-growing components of current expenditures, and to make progress on the implementation of long-standing tax policy reforms. Increasing domestic revenue as a percent of GDP by broadening the tax base and rationalizing fiscal exemptions is critical to avoid excessive expenditure compression.[8] Finally, the government is assumed to finance investment with relatively little recourse to domestic sources to support the exchange rate regime.

12. **The baseline scenario assumes that monetary policy will react promptly to stabilize reserves.** The central bank is assumed to tighten and rein in domestic credit growth to avoid reserve losses when necessary. Despite an increased vulnerability of the external accounts,[9] reserve coverage remains at a minimally satisfactory level and the baseline scenario assumes the peg is sustainable.

III. External Debt Sustainability

Baseline scenario

13. **The baseline scenario points to a deterioration of the external debt outlook in the next five years, followed by a gradual improvement.** The debt sustainability analysis uses policy-dependent external debt thresholds to gauge risks. Cape Verde is considered to have a strong policy performance, based on the latest World Bank's Country Policy and Institutional Assessment, thereby giving it slightly higher thresholds than countries with weaker policies. There is no significant or protracted breach of external debt indicators through 2031, but the present value (PV) of the debt-to-GDP approaches its policy-dependent threshold in 2016. The PV of debt-to-exports and the PV of debt-to-revenue remain well below their respective thresholds, reflecting the concessionality of outstanding debt and new external borrowing in the near-term.[10] The main message from the baseline scenario is broadly similar to the one from the previous DSA, despite differences in the debt profiles which are mostly explained by

[8] The World Bank and the IMF are working with the authorities on tax policy reform.

[9] International reserve coverage is projected to average 3.0 months of prospective imports between 2011 and 2015, against an average of 4.1 months projected in the previous DSA for the same period.

[10] Cape Verde lost a substantial amount of international reserves during 2011, which explains the large residual in Table 1. The authorities took corrective policy actions to tighten domestic macroeconomic policies, and counter the impact of the negative external environment.

an acceleration of the public investment program and improved measurement of the external debt.[11]

Box 1. Macroeconomic Assumptions for the Baseline Scenario

Real GDP growth experienced a solid recovery from the 2008 global crisis, remaining resilient until the first half of 2011, when growth started to slow down reflecting the troubled external environment. Hence, the medium-term outlook is weaker. Cape Verde's long-term growth prospects continue to be favorable, underpinned by a dynamic tourism sector and infrastructure investment.

Inflation is assumed to remain anchored to international levels given the peg to the euro, but also influenced by external volatility. Wage moderation and price flexibility, particularly in the tourism sector, will continue to facilitate adjustment to shocks.

External balance is restored. The current account is projected to remain negative during the DSA horizon due to the country's structural dependence on imports, but improve over time as exports become more diversified and import shares decline. Current transfers as a percent of GDP are expected to slowly decline as official grants are phased out and the domestic economy gradually outgrows remittance inflows. Capital account flows are expected to be modestly lower, particularly as the temporary increase in official disbursements to finance the public investment program comes to an end.

The overall fiscal balance is projected to improve, but remains a concern. After peaking in 2010 at double-digits, the fiscal deficit as a percentage of GDP is projected to decline and settle at an annual average of 3.3 percent in 2016-31. Tax revenue is expected to increase by about 4 percentage points of GDP over the DSA horizon due to the implementation of tax reforms and to the broadening of the tax base. External grants will decline over time, broadly offsetting tax revenue gains. Current expenditure is expected to grow in line with nominal GDP, while capital expenditure should reduce from its current high levels (18.7 percent of GDP in 2010) to less than 10.0 percent of GDP in the outer years.

Financing: The concessionality of new external loans will decline significantly in the medium term. The baseline assumes that this will happen around 2018–20 when concessional loans contracted up until 2013–15 should be close to full disbursement. Additionally, the baseline assumes that the authorities will continue to keep domestic borrowing low to support the peg.

Alternative scenarios and stress tests

14. **Bound tests and alternative scenarios show that Cape Verde's external debt is vulnerable to large shocks and a deteriorated economic environment (Table 2 and Figure 1).** The PV of external debt-to-GDP breaches the relevant thresholds under different stress tests. The exchange rate shock causes the largest and most protracted breach, highlighting the criticality of safeguarding the peg.[12] The export shock and the U.S. dollar GDP deflator shock also lead to breaches, though much less pronounced.[13] Additionally,

[11] In some cases, future disbursements of contracted debt were included as outstanding debt and/or had their repayment flows noted as service payments of the existing debt stock. Such disbursements were also recorded as external financing in the Medium-Term Fiscal Framework, leading to an overestimation of the debt stock and present value measures. The World Bank and IMF staff worked with the authorities during 2011 to assure consistency of public external debt data used in this DSA.

[12] A one-time 30 percent nominal depreciation relative to the baseline in 2012 is assumed.

[13] Export value growth at historical average minus one standard deviation in 2012–13; U.S. dollar GDP deflator at historical average minus one standard deviation in 2012–13.

under the assumption that new public sector loans are made on less favorable terms, the PV of external debt-to-GDP suffers a protracted breach of the relevant threshold.[14] Stress tests and less favorable assumptions provide some comfort about the repayment capacity of Cape Verde, as they do not imply any breach of the relevant thresholds for the PV of debt-to-exports and the PV of debt-to-revenue.

15. **A tail risk scenario resulting from severe disruption in the external environment could put significant pressure on debt indicators.** In the event of a strong deterioration of external conditions, Cape Verde could face tail risks involving a combination of negative shocks. Staff simulated the impact of an alternative scenario in which: (i) growth and foreign direct investment fall by 50.0 percent from their baseline projections for each year from 2012 to 2014; (ii) exports as a percentage of GDP fall by 10.0 percent from its baseline projection for each year from 2012 to 2014; and (iii) the interest rate on the marginal external debt needed to finance the balance of payments reaches five percent with a grace period of two years. The results indicate that the margin of comfort for some of the debt indicators under the baseline can be quickly eroded in this situation. Private sector external debt represents another source of vulnerability in such an extreme scenario, given potential higher interest costs associated with a higher country risk premium.

IV. Public Sector Debt Sustainability

Baseline scenario

16. **Total public debt in the baseline scenario reaches high levels and mirrors external debt dynamics.** Domestic debt is projected to remain manageable and to decline gradually over the medium term to support the exchange rate regime. However, public debt to GDP rises to high levels owing to large foreign-financed fiscal deficits, peaking at 87.4 percent in 2014 (Table 3). For the longer term, the domestic debt is projected to pick up slightly, remaining below 20.0 percent of GDP, as reliance on external financing declines more significantly.

Alternative scenarios and stress tests

17. **The debt outlook is vulnerable to the realization of losses on contingent liabilities associated with state-owned enterprises.** The latest government report on liabilities of selected state-owned enterprises covers six public companies.[15] The report indicates that underlying contingent liabilities associated to guaranteed and non-guaranteed debt reached 11.5 percent of GDP in 2010. Updated financial information on the debt of the loss-making

[14] This is an important alternative scenario to bear in mind given the existing concessionality of Portuguese credit lines, which rely on subsidies from that country's treasury.

[15] The Cape Verdean government holds a controlling stake in 13 financial and non-financial companies (more than 50.0 percent). The report covers key state-owned enterprises, but improving coverage is important to provide a more comprehensive assessment of risks.

TACV is lacking, which limits analysis. In any case, it is clear that both TACV and Electra represent a high fiscal risk to the budget. Both companies regularly need to access short-term credit facilities and resort to government guarantees. The government is committed to addressing the issues and the international community is providing support for restructuring.[16]

18. **Public debt sustainability indicators indicate vulnerability to alternative scenarios and stress tests (Table 4 and Figure 2).** However, the alternative scenario of an unchanged primary balance from 2011, which leads to a clearly unsustainable debt dynamics, is not a likely scenario since the scale-up of capital spending is expected to be temporary. Staff has also simulated the impact of a scenario of fiscal deterioration in which (i) domestic revenue as a percent of GDP remain stagnant at the 2011 level and (ii) current expenditure as a percent of GDP grows five percent faster than in the baseline projection for every year. The results indicate the critical importance of implementing the long-standing tax policy reforms and controlling growth of current expenditures.

V. CONCLUSIONS

19. **Staff assesses Cape Verde's debt as sustainable.** The debt indicators are projected to improve as the public investment program winds down. The long maturities and high concessionality of the existing debt mitigate rollover risks for now. Domestic revenue covers current expenditure, providing some room for adjustment in case there is an unexpected reduction of official grants or foreign financing. Cape Verde enjoys strong credibility due to its policies and democratic institutions, and continues to galvanize support from the international community. The tourism sector has remained resilient and competitive. International reserves are at minimally adequate levels and the authorities are committed to protecting the peg.

20. **The faster accumulation of external public debt in the midst of an uncertain global environment has increased risks around the baseline.** The prospects for the world economy have deteriorated over the last year. While Cape Verde has withstood well these tough times, staff believes that it is crucial to bolster progress on macroeconomic stability achieved so far. To that end, the authorities should seek faster debt reduction. This should involve moderation of the public investment pace, particularly domestically financed investment. Postponing new foreign-financed investments, especially those assessed to have more indefinite returns to the private sector, would also be prudent.[17] Such a strategy would also help to promote external balance and shore up international reserves by containing

[16] The World Bank expects to approve in 2012 a large energy loan to Electra (US$60 million) which will address priority investments and operational losses, and support the company's reform. The World Bank is also engaged with TACV.

[17] Public investment in Cape Verde as a percent of GDP (18.7 percent in 2010) was more than twice the sub-Saharan average (7.4 percent) and the highest among middle-income African economies (Botswana, Cape Verde, Ghana, Lesotho, Mauritius, Namibia, Senegal, Seychelles, South Africa, Swaziland, and Zambia).

growth of the import bill during uncertain times. Regarding debt management, the authorities should prioritize the development of analytical capacity in the Ministry of Finance to inform policymakers of risks around the baseline.

21. **The authorities view the rapid build-up of public debt as necessary to promote the country's development as a middle-income economy.** A key government objective is to create the foundations for economic diversification through public infrastructure investment. In their view, the temporary availability of concessional funds is a unique opportunity that cannot be missed. Staff understands the government's rationale, but it is of the view that public investment should be predicated on more conservative assumptions about the benefits of this investment, and thus undertaken more cautiously to match new external risk conditions and actual progress on structural reforms.[18] The staff's views and underlying assumptions of this DSA were discussed with the authorities.

[18] Slow progress on structural reforms has been holding back competitiveness for some time. There is general agreement about which reforms are needed: tax policy, revenue administration, state-owned enterprise restructuring (particularly of Electra and TACV), labor market flexibility, public investment management, and financial sector legislation. However, despite the commitment of the authorities, broader political traction has been lacking at times.

Table 1.Cape Verde: External Debt Sustainability Framework, Baseline Scenario, 2008-2031 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2008	2009	2010			2011	2012	2013	2014	2015	2016	2011-2016 Average	2021	2031	2017-203 Average
External debt (nominal) 1/	**68.1**	**68.4**	**70.8**			**70.4**	**72.5**	**75.1**	**75.8**	**74.6**	**73.4**		**59.2**	**35.5**	
o/w public and publicly guaranteed (PPG)	44.9	46.3	51.3			54.3	59.1	64.6	68.1	68.1	68.1		55.4	33.6	
Change in external debt	-5.6	0.3	2.3			-0.4	2.2	2.5	0.7	-1.2	-1.2		-3.2	-1.8	
Identified net debt-creating flows	-8.8	6.7	3.2			7.9	3.6	1.6	0.8	-0.1	-1.1		-1.8	0.4	
Non-interest current account deficit	**13.5**	**13.8**	**10.6**	**8.8**	**4.7**	**13.1**	**10.3**	**8.1**	**7.2**	**6.2**	**5.0**		**3.6**	**4.8**	4.0
Deficit in balance of goods and services	-123.7	-103.5	-105.5			-114.6	-111.9	-112.7	-113.4	-112.4	-111.3		-107.7	-97.9	
Exports	45.3	35.6	38.5			42.2	43.1	44.3	45.2	45.6	45.8		45.2	40.0	
Imports	-78.4	-67.9	-67.0			-72.4	-68.8	-68.4	-68.2	-66.8	-65.5		-62.5	-57.9	
Net current transfers (negative = inflow)	-20.3	-19.4	-20.4	-24.0	3.3	-18.7	-17.0	-17.4	-17.2	-16.4	-16.2		-14.7	-13.6	-14.3
o/w official	-6.7	-5.4	-6.4			-5.2	-2.5	-3.4	-3.5	-3.0	-3.0		-1.6	-0.5	
Other current account flows (negative = net inflow)	157.5	136.7	136.5			146.5	139.1	138.2	137.8	135.0	132.4		126.0	116.2	
Net FDI (negative = inflow)	**-13.4**	**-7.5**	**-6.7**	**-8.0**	**3.7**	**-4.1**	**-5.4**	**-5.1**	**-4.8**	**-4.5**	**-4.2**		**-4.1**	**-4.1**	-4.1
Endogenous debt dynamics 2/	**-8.9**	**0.4**	**-0.6**			**-1.1**	**-1.3**	**-1.5**	**-1.6**	**-1.8**	**-1.8**		**-1.3**	**-0.3**	
Contribution from nominal interest rate	2.2	1.9	1.8			2.0	2.0	2.0	2.1	2.0	2.0		1.9	1.6	
Contribution from real GDP growth	-3.9	-2.5	-3.6			-3.1	-3.3	-3.5	-3.7	-3.8	-3.8		-3.2	-1.9	
Contribution from price and exchange rate changes	-7.2	0.9	1.1			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**3.3**	**-6.5**	**-0.9**			**-8.3**	**-1.4**	**1.0**	**-0.1**	**-1.1**	**-0.1**		**-1.4**	**-2.2**	
o/w exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	55.2			54.3	55.2	56.4	56.4	55.6	55.0		48.0	31.7	
In percent of exports	...	...	143.5			128.7	128.0	127.2	124.8	121.9	120.2		106.1	79.4	
PV of PPG external debt	**...**	**...**	**35.8**			**38.2**	**41.8**	**45.9**	**48.7**	**49.2**	**49.7**		**44.2**	**29.8**	
In percent of exports	**...**	**...**	**93.0**			**90.6**	**96.9**	**103.6**	**107.7**	**107.8**	**108.6**		**97.8**	**74.5**	
In percent of government revenues	**...**	**...**	**165.0**			**170.4**	**186.8**	**199.7**	**208.3**	**206.4**	**204.9**		**178.7**	**114.1**	
Debt service-to-exports ratio (in percent)	**8.5**	**9.4**	**8.3**			**7.9**	**7.4**	**6.8**	**6.8**	**6.5**	**6.2**		**8.0**	**9.2**	
PPG debt service-to-exports ratio (in percent)	**4.6**	**5.6**	**4.7**			**4.8**	**4.4**	**4.0**	**4.2**	**3.9**	**3.7**		**5.9**	**7.5**	
PPG debt service-to-revenue ratio (in percent)	**7.5**	**8.5**	**8.4**			**9.0**	**8.5**	**7.8**	**8.1**	**7.5**	**7.0**		**10.7**	**11.5**	
Total gross financing need (Billions of U.S. dollars)	0.1	0.2	0.1			0.2	0.2	0.1	0.1	0.1	0.1		0.1	0.4	
Non-interest current account deficit that stabilizes debt ratio	19.0	13.5	8.2			13.5	8.1	5.6	6.5	7.4	6.2		6.8	6.6	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.2	3.7	5.4	6.1	2.0	5.0	5.1	5.2	5.3	5.4	5.5	5.3	5.5	5.6	5.5
GDP deflator in US dollar terms (change in percent)	10.8	-1.4	-1.6	5.8	8.5	10.7	3.3	2.1	2.0	2.8	2.7	3.9	2.7	2.8	2.7
Effective interest rate (percent) 5/	3.4	2.9	2.8	3.8	0.8	3.2	3.1	3.0	3.0	2.9	2.9	3.0	3.3	4.7	3.7
Growth of exports of G&S (US dollar terms, in percent)	24.6	-19.7	12.2	16.9	14.8	27.3	11.0	10.5	9.5	9.4	8.7	12.7	7.9	6.6	7.5
Growth of imports of G&S (US dollar terms, in percent)	18.6	-11.4	2.3	13.8	13.5	25.7	3.1	6.7	7.2	6.2	6.3	9.2	7.0	7.5	7.5
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	29.9	31.1	31.3	31.5	31.9	31.0	31.1	10.0	5.5	10.7
Government revenues (excluding grants, in percent of GDP)	27.9	23.4	21.7			22.4	22.4	23.0	23.4	23.8	24.3		24.7	26.1	25.2
Aid flows (in Billions of US dollars) 7/	0.2	0.2	0.3			0.2	0.1	0.1	0.2	0.1	0.1		0.1	0.1	
o/w Grants	0.1	0.1	0.1			0.1	0.1	0.1	0.1	0.1	0.1		0.1	0.1	
o/w Concessional loans	0.1	0.1	0.2			0.1	0.1	0.1	0.1	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			8.8	5.5	6.7	6.3	4.8	4.7		2.0	0.7	1.7
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			51.2	44.9	48.0	51.0	55.2	55.5		39.3	19.0	32.7
Memorandum items:															
Nominal GDP (Billions of US dollars)	1.6	1.6	1.7			1.9	2.1	2.3	2.4	2.6	2.8		4.3	9.6	
Nominal dollar GDP growth	17.7	2.3	3.7			16.3	8.6	7.4	7.4	8.4	8.4	9.4	8.4	8.5	8.4
PV of PPG external debt (in Billions of US dollars)			0.6			0.7	0.9	1.0	1.2	1.3	1.4		1.9	2.9	
(PVt-PVt-1)/GDPt-1 (in percent)						**8.9**	**6.8**	**7.5**	**6.4**	4.2	4.2	**6.3**	2.2	1.2	2.1

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2.Cape Verde: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2011-2031
(In percent)

				Projections				
	2011	2012	2013	2014	2015	2016	**2021**	2031
PV of debt-to GDP ratio								
Baseline	38	42	46	49	49	50	**44**	30
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	38	36	37	37	37	37	**34**	16
A2. New public sector loans on less favorable terms in 2011-2031 2	38	44	51	56	57	58	**55**	43
A3. Alternative Scenario : External tail risk	38	50	62	66	60	53	**48**	32
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	38	42	47	49	50	50	**45**	30
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	38	45	56	57	56	54	**44**	30
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	38	44	51	54	54	54	**49**	33
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	38	39	41	44	45	46	**44**	30
B5. Combination of B1-B4 using one-half standard deviation shocks	38	37	37	41	43	45	**45**	31
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	38	59	64	68	69	69	**62**	42
PV of debt-to-exports ratio								
Baseline	91	97	104	108	108	109	**98**	75
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	91	84	83	83	81	82	**76**	39
A2. New public sector loans on less favorable terms in 2011-2031 2	91	102	115	123	125	127	**122**	108
A3. Alternative Scenario : External tail risk	91	128	155	163	131	116	**105**	81
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	91	97	103	107	107	106	**98**	75
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	91	113	147	149	143	138	**115**	87
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	91	97	103	107	107	106	**98**	75
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	91	90	92	98	99	101	**98**	75
B5. Combination of B1-B4 using one-half standard deviation shocks	91	86	84	91	95	99	**99**	76
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	91	97	103	107	107	106	**98**	75
PV of debt-to-revenue ratio								
Baseline	170	187	200	208	206	205	**179**	114
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	170	163	161	160	155	154	**138**	59
A2. New public sector loans on less favorable terms in 2011-2031 2	170	197	221	238	239	239	**224**	166
A3. Alternative Scenario : External tail risk	171	222	269	284	250	219	**193**	123
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	170	188	203	212	208	205	**182**	116
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	170	201	242	246	233	221	**179**	114
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	170	198	222	231	227	224	**199**	127
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	170	175	177	189	190	191	**178**	114
B5. Combination of B1-B4 using one-half standard deviation shocks	170	166	162	177	182	187	**182**	117
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	170	262	280	293	287	283	**252**	161

Table 2.Cape Verde: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2011-2031 (continued)
(In percent)

Debt service-to-exports ratio								
Baseline	5	4	4	4	4	4	**6**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	5	4	3	2	1	1	**5**	5
A2. New public sector loans on less favorable terms in 2011-2031 2	5	4	4	5	5	5	**7**	10
A3. Alternative Scenario : External tail risk	5	5	6	22	19	18	**6**	8
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	5	4	4	4	4	4	**6**	8
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	5	5	5	7	10	9	**7**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	5	4	4	4	4	4	**6**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	5	4	4	3	2	2	**6**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	5	4	4	1	0	0	**6**	8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	5	4	4	4	4	4	**6**	8
Debt service-to-revenue ratio								
Baseline	9	8	8	8	8	7	**11**	11
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	9	8	7	4	2	1	**10**	7
A2. New public sector loans on less favorable terms in 2011-2031 2	9	8	8	9	9	9	**13**	15
A3. Alternative Scenario : External tail risk	9	9	10	38	37	34	**11**	12
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	9	9	8	8	8	7	**11**	12
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	9	8	8	12	16	15	**11**	11
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	9	9	9	9	8	8	**12**	13
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	9	8	7	5	3	3	**10**	11
B5. Combination of B1-B4 using one-half standard deviation shocks	9	9	7	3	0	0	**10**	12
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	9	12	11	11	11	10	**15**	16
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	0	0	0	0	0	0	**0**	0

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3.Cape Verde: Public Sector Debt Sustainability Framework, Baseline Scenario, 2008-2031

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2008	2009	2010	Average 5/	Standard Deviation 5/	2011	2012	2013	2014	2015	2016	2011-16 Average	2021	2031	2017-31 Average
Public sector debt 1/	67.9	68.8	74.9			77.7	82.1	85.8	87.4	84.8	82.3		67.1	49.5	
o/w foreign-currency denominated	44.9	46.3	51.3			54.3	59.1	64.6	68.1	68.1	68.1		55.4	33.6	
Change in public sector debt	-6.0	0.9	6.0			2.8	4.4	3.7	1.6	-2.6	-2.5		-2.4	-1.7	
Identified debt-creating flows	-2.9	-1.5	9.3			0.3	2.9	3.2	1.6	-2.5	-2.9		-2.4	-1.7	
Primary deficit	-0.3	4.8	9.0	3.3	3.0	8.1	6.6	7.0	5.6	1.9	1.4	5.1	1.4	0.4	1.2
Revenue and grants	33.3	28.8	28.0			27.6	24.8	26.4	26.9	26.8	27.3		26.4	26.6	
of which: grants	5.5	5.4	6.3			5.2	2.5	3.4	3.5	3.0	3.0		1.6	0.5	
Primary (noninterest) expenditure	33.0	33.5	37.0			35.7	31.4	33.3	32.5	28.7	28.6		27.7	27.1	
Automatic debt dynamics	-1.9	-6.4	0.4			-7.7	-3.7	-3.9	-4.1	-4.4	-4.4		-3.8	-2.1	
Contribution from interest rate/growth differential	-4.5	-2.3	-3.2			-3.5	-3.0	-3.3	-3.5	-4.0	-4.1		-3.1	-1.7	
of which: contribution from average real interest rate	-0.1	0.2	0.3			0.1	0.8	0.8	0.9	0.5	0.3		0.5	1.0	
of which: contribution from real GDP growth	-4.3	-2.4	-3.5			-3.6	-3.8	-4.1	-4.3	-4.5	-4.4		-3.6	-2.7	
Contribution from real exchange rate depreciation	2.6	-4.1	3.6			-4.1	-0.7	-0.6	-0.7	-0.5	-0.3		...	...	
Other identified debt-creating flows	-0.7	0.1	-0.1			-0.1	0.0	0.1	0.1	0.1	0.0		0.0	0.0	
Privatization receipts (negative)	-0.7	0.1	-0.1			-0.1	0.0	0.1	0.1	0.1	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-3.1	2.5	-3.3			2.5	1.5	0.6	0.0	-0.1	0.5		0.0	0.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	59.4			61.6	64.7	67.1	68.0	65.9	64.0		55.9	45.8	
o/w foreign-currency denominated	...	...	35.8			38.2	41.8	45.9	48.7	49.2	49.7		44.2	29.8	
o/w external	...	...	35.8			38.2	41.8	45.9	48.7	49.2	49.7		44.2	29.8	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	2.9	7.8	11.9			11.3	9.7	9.9	8.6	4.6	3.9		4.7	4.4	
PV of public sector debt-to-revenue and grants ratio (in percent)	…	…	212.3			223.1	260.8	254.5	252.7	245.6	234.6		212.0	171.8	
PV of public sector debt-to-revenue ratio (in percent)	…	…	273.5			274.8	289.5	291.9	290.9	276.3	263.6		226.0	175.3	
o/w external 3/	…	…	165.0			170.4	186.8	199.7	208.3	206.4	204.9		178.7	114.1	
Debt service-to-revenue and grants ratio (in percent) 4/	9.7	10.5	10.4			11.8	12.6	11.3	11.0	10.3	9.2		12.7	14.8	
Debt service-to-revenue ratio (in percent) 4/	11.6	12.9	13.3			14.6	14.0	12.9	12.7	11.6	10.4		13.5	15.1	
Primary deficit that stabilizes the debt-to-GDP ratio	5.7	3.8	3.0			5.3	2.2	3.2	4.0	4.5	3.9		3.8	2.1	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	6.2	3.7	5.4	6.1	2.0	5.0	5.1	5.2	5.3	5.4	5.5	5.3	5.5	5.6	5.5
Average nominal interest rate on forex debt (in percent)	1.1	1.1	1.1	1.2	0.2	1.4	1.5	1.4	1.5	1.3	1.3	1.4	1.7	2.9	2.1
Average real interest rate on domestic debt (in percent)	1.2	0.6	1.8	3.3	2.6	1.8	2.3	2.6	2.6	2.5	2.6	2.4	3.7	3.6	3.5
Real exchange rate depreciation (in percent, + indicates depreciation)	6.0	-9.5	8.1	-2.3	12.3	-8.5	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	3.2	4.2	3.3	1.8	2.1	3.9	3.4	2.9	2.9	2.8	2.7	3.1	2.7	2.8	2.7
Growth of real primary spending (deflated by GDP deflator, in percent)	0.1	0.1	0.2	0.1	0.1	0.0	-0.1	0.1	0.0	-0.1	0.1	0.0	0.0	0.1	0.1
Grant element of new external borrowing (in percent)	...	...	...	…	…	29.9	31.1	31.3	31.5	31.9	31.0	31.1	10.0	5.5	...

Sources: Country authorities; and staff estimates and projections.

1/ Central government and publically-guaranteed debt.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

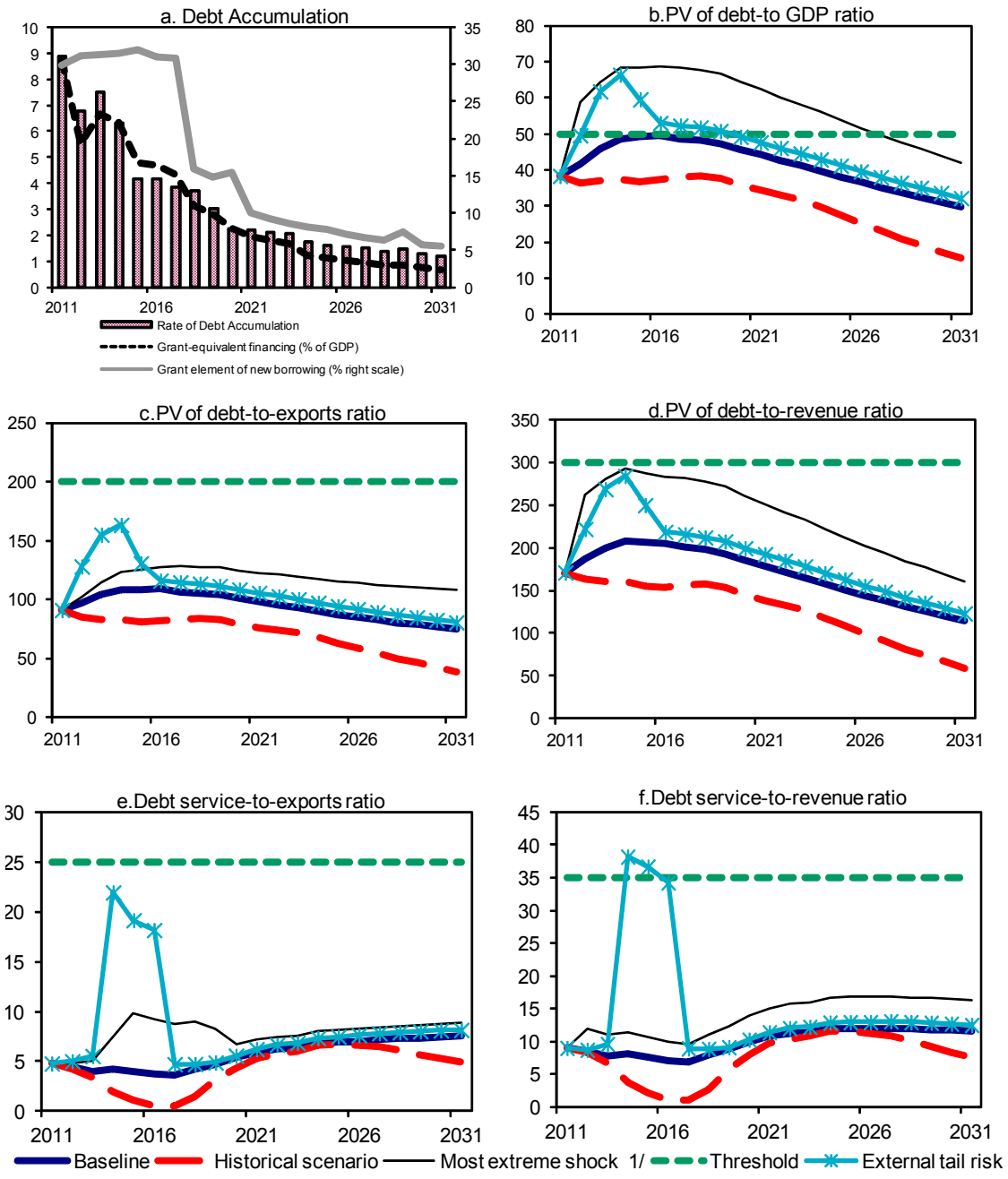
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4.Cape Verde: Sensitivity Analysis for Key Indicators of Public Debt 2011-2031

	Projections							
	2011	2012	2013	2014	2015	2016	2021	2031
PV of Debt-to-GDP Ratio								
Baseline	62	65	67	68	66	64	56	46
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	62	61	59	57	57	57	57	67
A2. Primary balance is unchanged from 2011	62	66	70	73	78	83	105	152
A3. Permanently lower GDP growth 1/	62	65	68	69	68	67	64	71
A4. Alternative Scenario: Fiscal deterioration	62	65	69	71	70	70	69	73
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	62	66	69	71	69	67	61	55
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	62	64	66	67	65	63	55	45
B3. Combination of B1-B2 using one half standard deviation shocks	62	63	63	64	62	60	53	43
B4. One-time 30 percent real depreciation in 2012	62	80	80	80	77	75	66	60
B5. 10 percent of GDP increase in other debt-creating flows in 2012	62	75	78	78	76	74	65	53
PV of Debt-to-Revenue Ratio 2/								
Baseline	223	261	254	253	246	235	212	172
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	223	245	224	213	212	208	219	253
A2. Primary balance is unchanged from 2011	223	267	265	272	290	303	399	571
A3. Permanently lower GDP growth 1/	223	262	257	258	253	244	241	265
A4. Alternative Scenario: Fiscal deterioration	223	263	268	274	277	276	288	319
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	223	264	262	262	256	247	233	207
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	223	259	251	249	242	231	209	169
B3. Combination of B1-B2 using one half standard deviation shocks	223	253	239	238	231	221	200	163
B4. One-time 30 percent real depreciation in 2012	223	321	305	297	288	275	252	224
B5. 10 percent of GDP increase in other debt-creating flows in 2012	223	304	294	290	283	271	245	198
Debt Service-to-Revenue Ratio 2/								
Baseline	12	13	11	11	10	9	13	15
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	12	13	10	9	5	1	12	33
A2. Primary balance is unchanged from 2011	12	13	12	12	13	14	47	103
A3. Permanently lower GDP growth 1/	12	13	11	11	11	10	16	32
A4. Alternative Scenario: Fiscal deterioration	12	13	12	12	11	10	15	21
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	12	13	11	11	11	10	16	22
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	12	13	11	11	10	8	12	14
B3. Combination of B1-B2 using one half standard deviation shocks	12	13	11	10	8	5	9	12
B4. One-time 30 percent real depreciation in 2012	12	14	14	14	14	14	23	35
B5. 10 percent of GDP increase in other debt-creating flows in 2012	12	13	13	13	22	21	21	22

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

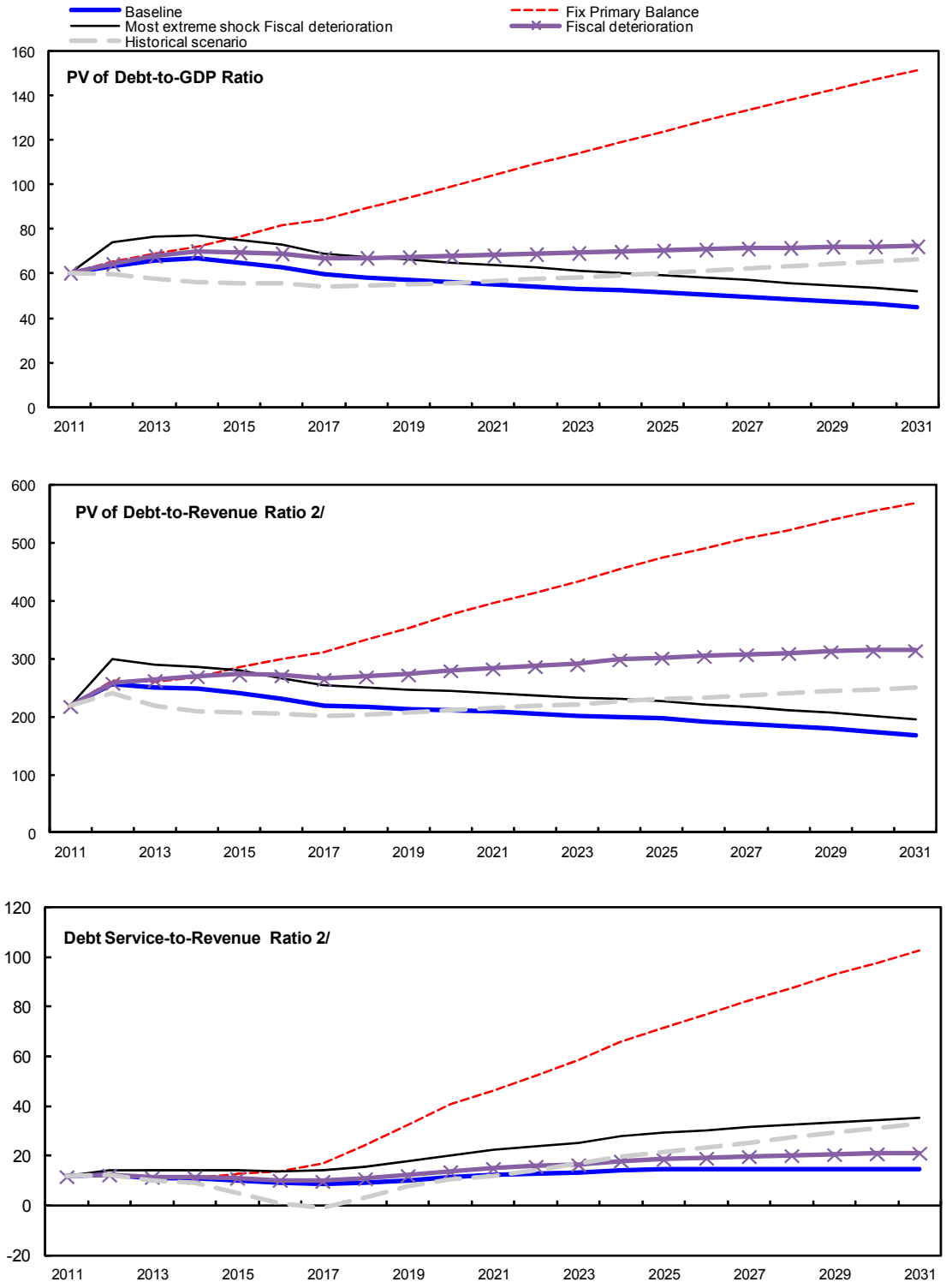
Figure 1. Cape Verde: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2011-2031 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2021. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2.Cape Verde: Indicators of Public Debt Under Alternative Scenarios, 2011-2031 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2021.

2/ Revenues are defined inclusive of grants.